

Prolog Program for PLANETS DB

1. Aim

To write and execute a Prolog program to create a database of the planets in the solar system and retrieve information such as position and type using facts, rules, and queries.

2. Objective

- To understand database representation in Prolog using facts.
- To represent each planet's orbital position and type (terrestrial, gas giant, ice giant).
- To retrieve planet information using rules and queries.
- To filter facts based on a shared attribute (e.g. all gas giants).

3. Algorithm

1. Start the program.
2. Create a Prolog database using the predicate planet/3.
3. Store each planet's name, position from the sun, and type as facts.
4. Define a rule find_planet/3 to retrieve a planet's details.
5. Enter the planet's name (or type) as a query.
6. Search the planet database for the given name or type.
7. If a match exists, display the planet's details; otherwise Prolog returns false.
8. Stop the program.

4. Flowchart



END

5. Code with Output

Code:

```
% -----  
% PLANETS Database  
% planet(Name, Position, Type)  
% -----  
planet(mercury, 1, terrestrial).  
planet(venus, 2, terrestrial).  
planet(earth, 3, terrestrial).  
planet(mars, 4, terrestrial).  
planet(jupiter, 5, gas_giant).  
planet(saturn, 6, gas_giant).  
planet(uranus, 7, ice_giant).  
planet(neptune, 8, ice_giant).  
  
% -----  
% Rule to find a planet's details  
% -----  
find_planet(Name, Position, Type) :-  
    planet(Name, Position, Type).  
  
% Rule to list all planets of a given type  
list_by_type(Type) :-  
    planet(Name, _, Type),  
    format("~w~n", [Name]),  
    fail.  
list_by_type(_).
```

Output:

```
?- find_planet(earth, P, T).  
P = 3,  
T = terrestrial.  
  
?- list_by_type(gas_giant).  
jupiter  
saturn  
true.
```

6. Result

The Prolog program was successfully implemented to create a planets database and retrieve orbital position and type information using facts and queries.

7. Conclusion

Thus, the PLANETS database was successfully implemented using Prolog. The experiment demonstrates how facts can model a fixed real-world dataset and how queries can both look up single records and filter records sharing a common attribute.